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Rene-Deskeptic: Engineering Metacognitive Doubt to Eliminate LLM Hallucinations

The Problem: Parametric Stubbornness Current large language models (LLMs) are trained to optimize associative pattern matching, making them consensus engines rather than reasoning engines. When provided an LLM with new context—such as a specific book or document—they consistently ignore the supplied text and switch to their memorized patterns from pretraining. In controlled tests, Gemini 3.1 Pro was given a PDF of a book and asked for key quotes, Gemini repeatedly cited the same famous quotes it knew from training, bypassing the provided document entirely. This parametric stubbornness undermines trust, amplifies hallucinations, and renders chain-of-thought reasoning meaningless—models simply construct probabilistic defenses of their training biases.

The Solution: A Skeptical Architecture Our proposed architecture, Rene-Deskeptic, forces LLMs to doubt their own knowledge and forensically verify against provided context. It rests on three pillars: Retro-Search Distillation Prunes reasoning traces from teacher models (e.g., DeepSeek-R1) to remove overthinking and preserve only efficient, correct paths. Prior work shows this reduces reasoning length by 31.2% while improving accuracy by 7.7% on complex tasks. Epistemic Regret Minimization (ERM) Penalizes the model for being right for the wrong reasons by maintaining an explicit causal graph of beliefs. Loss function combines task success with causal accuracy, preventing spurious correlations from being reinforced.

Adversarial Self-Critique (ASC) & Deep Truth Mode At inference, the model generates multiple parallel reasoning tracks (e.g., using only primary sources, using pure logic) and then adopts a hostile critic persona to attack all tracks. Only those that survive the “crucifixion round” are output, along with explicit falsification pathways. These components are designed to be model-agnostic and can be integrated into any frontier-scale foundation model.

Expected Impact Hallucination reduction: Near-elimination of context-ignoring behavior. Reasoning efficiency: 30% shorter chains with higher accuracy. Generalization: Demonstrable improvement in code generation, chess strategy, and medical fact verification. Benchmark targets: >1977 Elo in chess, >90% pass@1 on LiveCodeBench, >94% rejection of medical sycophancy prompts.

Current Status & Resource Needs: Rene-Deskeptic is currently in advanced architectural design, with detailed mathematical specifications and simulation results. The next phase requires: GPU compute for fine-tuning a large-scale MoE foundation model. Mentorship from academic or industry researchers in causal inference, LLM alignment, or cognitive architectures. Collaboration on rigorous evaluation benchmarks and ablation studies. We are seeking partnerships with universities and industry labs to accelerate development.

Why Now? The window for advances in AI reasoning is open. By integrating Rene-Deskeptic directly into the workspace, we can access paths beyond statistical parroting toward genuine understanding. Your support can help turn this blueprint into a working system that sets a new standard for the industry of AI.

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Works Cited

- Chang, Edward Y., and Stanford University. “Right for the Wrong Reasons: Epistemic Regret Minimization for Causal Rung Collapse in LLMs.” *arxiv*, Edward Y. Chang, 12 February 2026, <https://arxiv.org/html/2602.11675v1>.
- Lu, Ximing, et al. “Retro-Search: Exploring Untaken Paths for Deeper and Efficient Reasoning.” *Hugging Face*, Ximing Lu, 6 April 2025, <https://arxiv.org/abs/2504.04383>.
- Zhang, Jue, et al. “From Reasoning to Answer: Empirical, Attention-Based and Mechanistic Insights into Distilled DeepSeek R1 Models.” *Aclanthology*, Zhang et al, 2025, <https://aclanthology.org/2025.emnlp-main.198/>.